

SOA Lab Test – Problem Statements

Exercise 1: REST Service + Client + Testing

Problem 1

Design a **User Management REST Service** that allows creating and retrieving users.

Demonstrate the service using a client script.

Problem 2

Build a **Product Inventory Service** with full CRUD operations.

Test all endpoints using Postman or curl.

Problem 3

Develop a **Student Record Service** that stores student details.

Ensure the service supports GET and POST operations with JSON responses.

Problem 4

Create a **Book Management API** for a library system.

Implement update and delete operations and verify functionality using a client.

Problem 5

Implement an **Order Service** and generate API documentation (Swagger/OpenAPI or manual).

Explain how another client can use the service.

Exercise 2: Messaging + Loose Coupling (Event-Driven SOA)

Problem 6

Design an **Order Event System** where a publisher sends a message whenever an order is placed.

A consumer processes the order independently.

Problem 7

Create a **User Notification System** where events like "User Registered" are published and consumed by a notification service.

Problem 8

Develop a **Payment Processing Event System** where transaction events are published and logged by another service.

Problem 9

Implement a **Log Aggregation System** where multiple services publish logs and a consumer collects and processes them.

Problem 10

Build an **Inventory Update System** where stock updates are published and consumed by a separate service.
Ensure no direct dependency between services.

Exercise 3: Serverless + SOA Integration

Problem 11

Create a **Text Processing Serverless Function** that converts input text to uppercase and expose it as a REST endpoint.

Problem 12

Develop a **Simple Calculator Function** using serverless computing that performs addition and multiplication based on input.

Problem 13

Design an **Image Processing Flow** where a request triggers a serverless function to process an image and return output.

Problem 14

Build a **File Metadata Service** where uploading a file triggers a serverless function to extract details (size/type).

Problem 15

Implement a **Workflow Integration System** where a REST service invokes a serverless function as part of processing.

Exercise 4: AI/ML Service in SOA

Problem 16

Develop a **Sentiment Analysis Service** that classifies text as positive or negative and expose it via REST.

Problem 17

Create an **Image Classification Service** using a pre-trained model and return predicted labels.

Problem 18

Build a **Spam Detection Service** that analyzes input messages and classifies them as spam or not.